Theory of Value

An Axiomatic Analysis of Economic Equilibrium
Gerard Debreu (1959)

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October 1, 2022

Chapter 1: Mathematics

01. Introduction
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05. Real Numbers
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07. Continuous Functions
08. Continuous Correspondences
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10. Fixed Points

Fixed Points

a. Consider a set S and a function f from S to S, i.e., a transformation of S into itself. Great interest is attached to the existence of an element x' such that $x' = f(x')$, i.e., which coincides with its image, or which does not move in the transformation. Such an element is called a fixed point of the transformation f (see fig. 9.a).

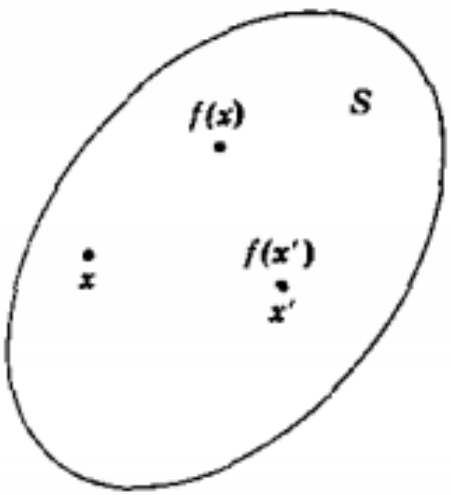


Fig. 9.a

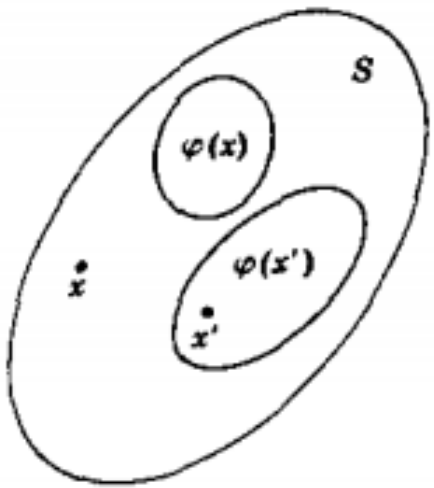


Fig. 9.b

Fixed Points

b. One can prove the fundamental theorem:

(1) (Brouwer's Theorem) If S is a non-empty, compact, convex subset of $\mathbb{R}^N$, and if f is a continuous function from S to S, then f has a fixed point.

Fixed Points

(0) (Brouwer's Theorem) Let $D = \{x \in \mathbb{R}^N \mid \|x\| \leqslant 1\}$. If function $h : D \to D$ is continuous, then h has a fixed point.

Define a homeomorphic mapping $\phi : D \to S$, we prove that (0) and (1) are equivalent.

(0) $\Longrightarrow$ (1): Define function $h = \phi^{-1} \circ f \circ \phi : D \to D$, then h is continuous and has a fixed point x_0. That is, $h(x_0) = x_0$ or $\phi^{-1}(f(\phi(x_0))) = (\phi^{-1} \circ f \circ \phi)(x_0) = x_0$, then $f(\phi(x_0)) = \phi(x_0)$ and hence $\phi(x_0)$ is a fixed point of f.

(1) $\Longrightarrow$ (0): Define function $f = \phi \circ h \circ \phi^{-1} : S \to S$, then f is continuous and has a fixed point x_0. That is, $f(x_0) = x_0$ or $\phi(f(\phi^{-1}(x_0))) = (\phi \circ h \circ \phi^{-1})(x_0) = x_0$, then $h(\phi^{-1}(x_0)) = \phi^{-1}(x_0)$ and hence $\phi^{-1}(x_0)$ is a fixed point of h.

Fixed Points

(0) (Brouwer's Theorem) Let $D^N = \{x \in \mathbb{R}^N \mid \|x\| \leqslant 1\}$. If function $h : D^N \to D^N$ is continuous, then h has a fixed point.

(0) (Brouwer's Theorem) Let $D = \{x \in \mathbb{R} \mid \|x\| \leqslant 1\} = [-1, 1]$. If function $h : D \to D$ is continuous, then h has a fixed point.

Define function $h' = h - Id : D \to D$, then h' is continuous with $h'(x) = h(x) - x$ and hence $h'(-1) = h(-1) + 1 \geqslant 0$ and $h'(1) = h(1) - 1 \leqslant 0$, thus h' has a solution to $h'(x) = 0$, that is, h has a fixed point.

Fixed Points

c. The generalization of this result to correspondences from a set to itself will play an essential role later on. Consider now a set S and a correspondence φ from S to S. A fixed point of the correspondence φ is an element x' such that $x' \in \varphi(x')$, i.e., which belongs to its image-set (see fig. 9.b).

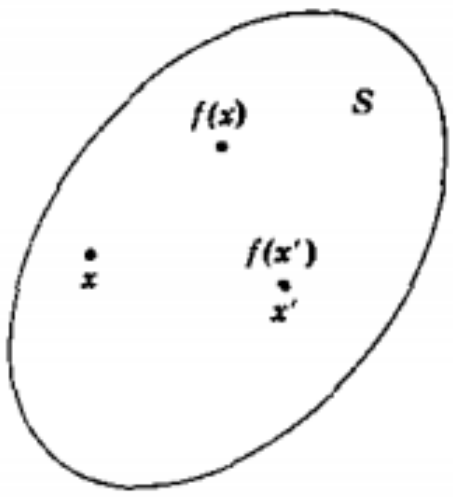


Fig. 9.a

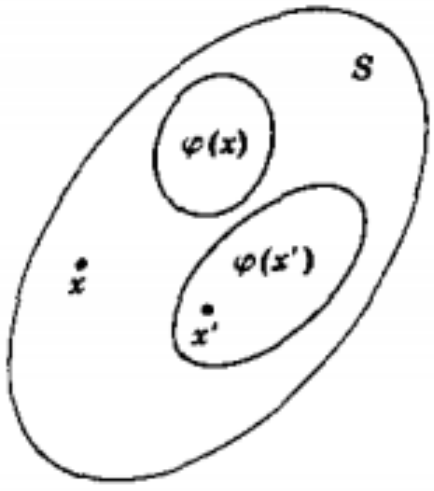


Fig. 9.b

Fixed Points

d. One can prove:

(2) (Kakutani's Theorem) If S is a non-empty, compact, convex subset of $\mathbb{R}^N$, and if φ is a upper semi-continuous correspondence from S to S such that for all $x \in S$ the set $\varphi(x)$ is convex (non-empty), then φ has a fixed point.